

Reversed and Non-Equivalent Ratios

How to work:

- A ratio is written in the order the question asks for. “ A to B ” means A on top, B on the bottom — reversing it names a different comparison.
- Two ratios are **equivalent** when you can get one from the other by multiplying (or dividing) *both* parts by the same number. Adding the same number to both parts does not work.
- To test equivalence, simplify both ratios or compare their unit rates.
- Show your work in the space, then write your answer on the line.

1. A muffin recipe uses 3 cups of flour for every 2 cups of sugar. Write the ratio of flour to sugar as a fraction in simplest form.

Answer: _____

2. Are the ratios 4 : 6 and 6 : 9 equivalent? Simplify both, then write = or $\neq$ between them and say which it is.

$$\frac{4}{6} \text{ ————— } \frac{6}{9}$$

Answer: _____

3. Find the mistake. Ms. Ruiz's class has 12 boys and 18 girls. Leo is asked for the ratio of **girls to boys** and writes $\frac{12}{18}$.

- (a) Explain in one sentence why the order changes what the ratio means.
- (b) Write the ratio of girls to boys correctly, in simplest form.

Answer: _____

4. A store sells 3 pencils for 2 dollars. Complete the equivalent-ratio table, then find the cost x of 12 pencils.

pencils	3	12
dollars	2	x

Answer: _____

5. Find the mistake. A drink mix uses 4 cups of water to 6 scoops of powder. Mia needs 6 cups of water instead. She adds 2 to each part and writes 8 scoops of powder.

- (a) Explain why adding the same number to both parts does not keep a ratio equivalent.
- (b) Find the number of scoops that really is equivalent to 4 cups : 6 scoops.

Answer: _____

6. One car travels 150 miles on 5 gallons of gas. Another travels 210 miles on 7 gallons. Find each unit rate in miles per gallon, then write $=$ or $\neq$ between the two rates to say whether they are equivalent.

Answer: _____

7. Find the mistake. A printer prints 24 pages in 3 minutes. Sam reports the rate as 0.125 pages per minute.

- (a) Explain in one sentence how to tell which number belongs on top in a rate.
- (b) Find the correct rate in pages per minute.

Answer: _____

8. Challenge. One row of this ratio table does not belong.

blue paint	2	4	6	8
white paint	5	10	15	22

- (a) Compare the last column with $2 : 5$ by writing $<$, $=$, or $>$ between $\frac{8}{22}$ and $\frac{2}{5}$.
- (b) Replace the wrong entry with the number that makes the column equivalent to $2 : 5$.

Answer: _____